Monthly Intelligence Report

Edition 001 — The Dublin–Cork Grid Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

994

319 HV · 675 MV

GRID LINES MAPPED

2,306

transmission + distribution

MC SIMULATIONS

9.9 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 10,000+ source data points per run, executes 9.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 223 million arithmetic operations — producing 59,243 output data points with full uncertainty quantification across 994 substations and 2,306 power lines.

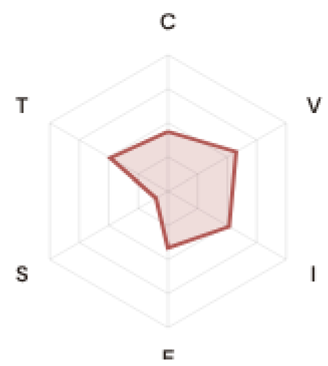
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Ireland's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

North Quays 110kV Substation

Leinster, Dublin · 110 kV

0.454	Medium	0.068	83th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- C Continuity: 0.436 / 0.30 (145%)
- I Infrastructure: 0.517 / 0.25 (207%)
- S Saturation: 0.100 / 0.20 (50%)
- V Voltage: 0.576 / 0.10 (576%)
- E Economic: 0.415 / 0.10 (415%)
- T Transition: 0.487 / 0.05 (974%)

MODIFIERS

- **R3 Consequence:** 1.117 ↑ amplifies
- **R4 Graph Criticality:** 0.931 ↓ dampens
- **R6a Restoration:** 1.011 ↑ amplifies
- **R6b Seismic:** 1.070 ↑ amplifies
- **R7 Digital Readiness:** 1.001 → neutral

This month's spotlight — automatically selected for April 2026 — is **North Quays 110kV Substation** in Leinster, Dublin. With an R_final of 0.454 (Medium band, 83th fleet percentile), its risk profile is dominated by the T (Transition) component at 974% of its weight ceiling, followed by V (Voltage) at 576%. Modifiers collectively shift R_base by 12.6%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Ireland's National Energy & Climate Plan (NECP) and Climate Action Plan digitalisation priorities require.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

33

High/Critical + R7 < median

AVG R7 MODIFIER

0.9797

Range [0.99, 1.05]

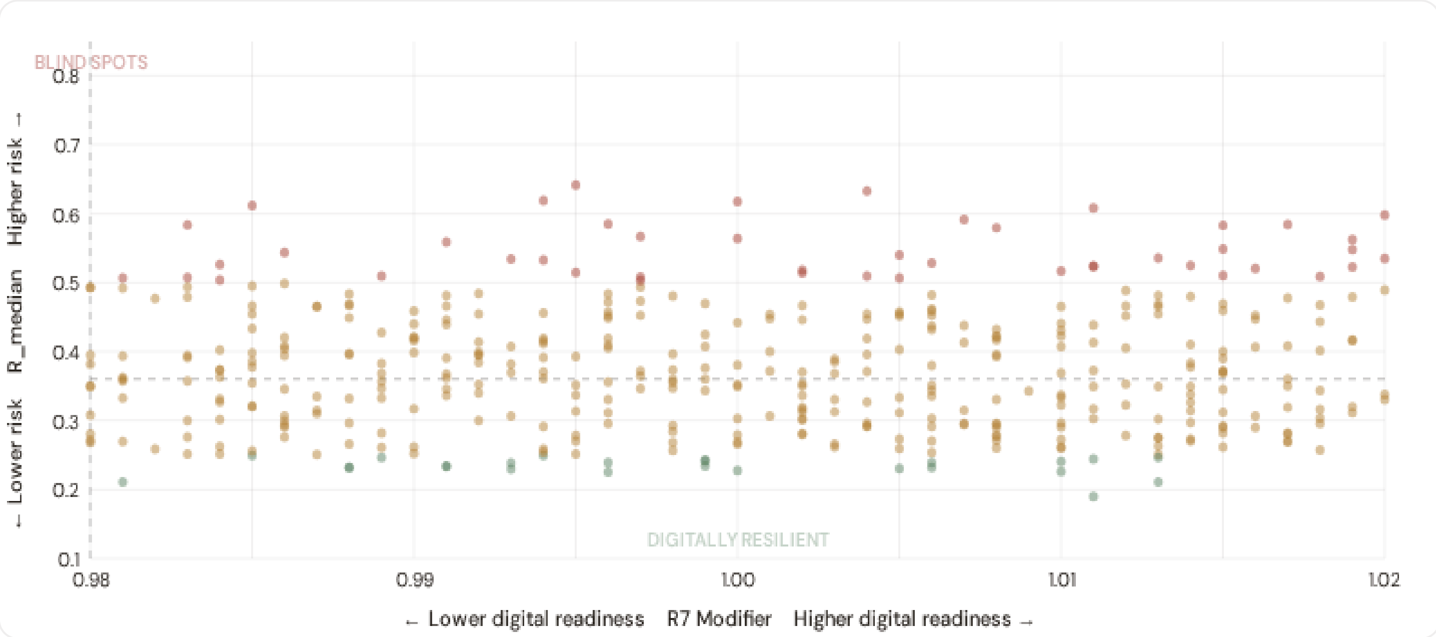
HIGH-CONSEQUENCE SUBS

425

42.8% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **33 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9800). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Dublin (33)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: €15–30/kWh

Heavy industry, petrochemical, large manufacturing — highest economic loss per interruption

307 substations (30.9%) High/Critical: **72** (23.5%) Avg R: **0.448** Avg E: **0.362** / 0.10

Low 0 Med 235 High 72 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: €8–15/kWh

Diversified industry, logistics, food processing — significant continuity requirements

199 substations (20.0%) High/Critical: **13** (6.5%) Avg R: **0.400** Avg E: **0.353** / 0.10

Low 0 Med 186 High 13 Crit 0

🏪 Commercial / SME-Dense

Est. VoLL: €3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

145 substations (14.6%) High/Critical: **3** (2.1%) Avg R: **0.356** Avg E: **0.330** / 0.10

Low 2 Med 140 High 3 Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: €1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

343 substations (34.5%) High/Critical: **0** (0.0%) Avg R: **0.285** Avg E: **0.311** / 0.10

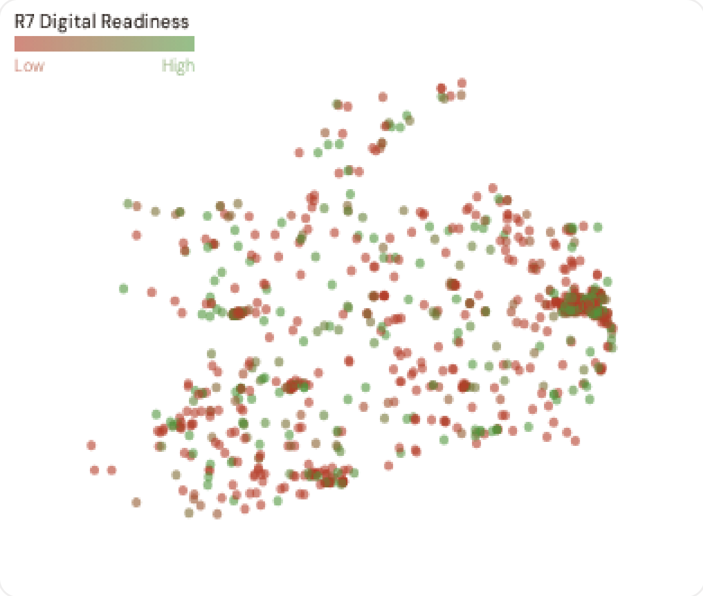
Low 65 Med 278 High 0 Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate for Ireland places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **72 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Dublin (222), Cork (44), Limerick (17)** — regions where industrial corridors depend on uninterrupted power for continuous processes (steel, glass, automotive). A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 7.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 County Digital Readiness Ranking

Counties ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify counties combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Leinster ⚠ high R	0.9786
2	Munster	0.9791
3	Ulster	0.9799
4	Connacht	0.9843

County-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Leinster** has the lowest average R7 (0.9786), while **Connacht** leads at 0.9843 — a spread of 0.0057 across the R7 range. **1 counties** combine below-median digital readiness with above-median grid risk, making them priority targets for Climate Action Plan / NECP digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9797, median = 0.9800; 0 substations classified HIGH cyber-exposure; 33 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging counties where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES	WEEKLY / MONTHLY	QUARTERLY / ANNUAL
35	5	22
95 variables	High-frequency feeds	Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

Data Source Registry — April 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
EirGrid-N	EirGrid TDP (Transmission Development Plan)	MONTHLY	TSO zone	2 vars
DREAL	EPA (Environmental Protection Agency)	CONTINUOUS	County	1 var
ODRE	ESBN Open Data (ESB Networks)	QUARTERLY	County	5 vars
SEAI	SEAI EV Data	QUARTERLY	County	1 var
BPI	Enterprise Ireland Innovation	QUARTERLY	County	1 var
CRU	CRU (Commission for Regulation of Utilities)	ANNUAL	County	8 vars
CSO	CSO (Central Statistics Office)	ANNUAL	County	8 vars
SEAI	SEAI (Sustainable Energy Authority Ireland)	ANNUAL	County	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MTEF	Department of Environment, Climate and Communications	ANNUAL	Province	2 vars
CEREMA	Cerema (Centre d'études risques)	ANNUAL	County	1 var
Revenue	Revenue Commissioners Statistics	ANNUAL	County	1 var
RPII	RPII Nuclear Safety	ANNUAL	County	1 var
CSO-Poverty	CSO Poverty Statistics	ANNUAL	County	2 vars
ESBN	ESB Networks Open Data	ANNUAL	County	2 vars
NESC	NESC (National Economic and Social Council)	ANNUAL	County	2 vars
CRU-M	CRU Annual Report	ANNUAL	DSO-level	2 vars
CSO-Census	CSO Census Geography	ANNUAL	County	1 var
ESRI	ESRI (Economic and Social Research Institute)	ANNUAL	Province	1 var
SEAI-R	SEAI Regions	ANNUAL	Province	1 var
BRGM	GSI (Geological Survey Ireland)	STATIC	County	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars

DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	CSO County Registry	STATIC	County	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	County	1 var
EirGrid	EirGrid Data Portal	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF-OM	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-OM	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

Monthly (3 sources) — Grid load + procurement data

ESBN Registry · EirGrid éCO2mix · EirGrid SDDR

Quarterly (2 sources) — DER registry + business data + macro indicators

SEAI · Enterprise Ireland

Annual (20 sources) — Core grid quality + demographics + environment

CRU · CSO · CSO-Poverty · ESBN · MTEF · SEAI · Revenue · EEA · Eurostat · DESI · NESC · CRU-M · CSO-Census · ESRI · SEAI-R · RPII · Open-Meteo · JRC · DREAL · OPW

Static (8 sources) — Standards + scenarios + seismic hazard

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · BRGM · ISO 9223 · COG/INSEE · ENTSO-E

C.2 Fleet Score Snapshot

FLEET SIZE
994
substations scored

MEDIAN R
0.360
P5=0.239 · P95=0.529

HIGH + CRITICAL
88
8.9% of fleet

HIGH CONFIDENCE
100%

994 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 67 (6.7%) · Medium 839 (84.4%) · High 88 (8.9%) · Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **994 substation scores remain unchanged** this edition. The fleet median R of 0.360 (mean 0.369) reflects a distribution skewed toward the lower bands, with 6.7% in Low and 84.4% in Medium. The first material score changes are expected when CRU publishes the annual CRU-Quality indicators (affecting C1–C4 and R6a) and when ESBN-Data refreshes the DER installation register (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	§2, §4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	§5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	§5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	§5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	§5
F6	ENHANCED	R7 Digital Readiness — County-level DESI model	§5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	§2, §8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	§6
L3	ENHANCED	V_socio Fiscal Enrichment — DGFIP + ANCT + energy price	§5
L4	ENHANCED	R3 Demographic Shift Amplifier — INSEE net migration	§5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	§8, §12
G1	DATA	d02 ESBN upgraded to quarterly registry — County-level DER registry	§12
G2	DATA	d05 BRGM upgraded to live API — County-level geological data	§12
G3	DATA	d06 OSM upgraded to Overpass API — 994 real substations	§12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	§5
G5	NEW	Network & Topology layer (H): 7 variables — EirGrid TDP (Transmission Development Plan), ring analysis	§12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	§12

SECTION D — MONTHLY DEEP-DIVE

The Dublin–Cork Grid Corridor: Where Wind Integration Meets Grid Modernisation Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Dublin–Cork corridor · **002** Wind integration hotspots (Munster/Connacht) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Data centre demand and grid capacity risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Dublin–Cork Corridor, Why Now

DUBLIN–CORK CORRIDOR SUBSTATIONS

549

35 EHV · 514 HV

HIGH BAND

87

15.8% of corridor

CORRIDOR MEDIAN R

0.405

Fleet median: 0.360

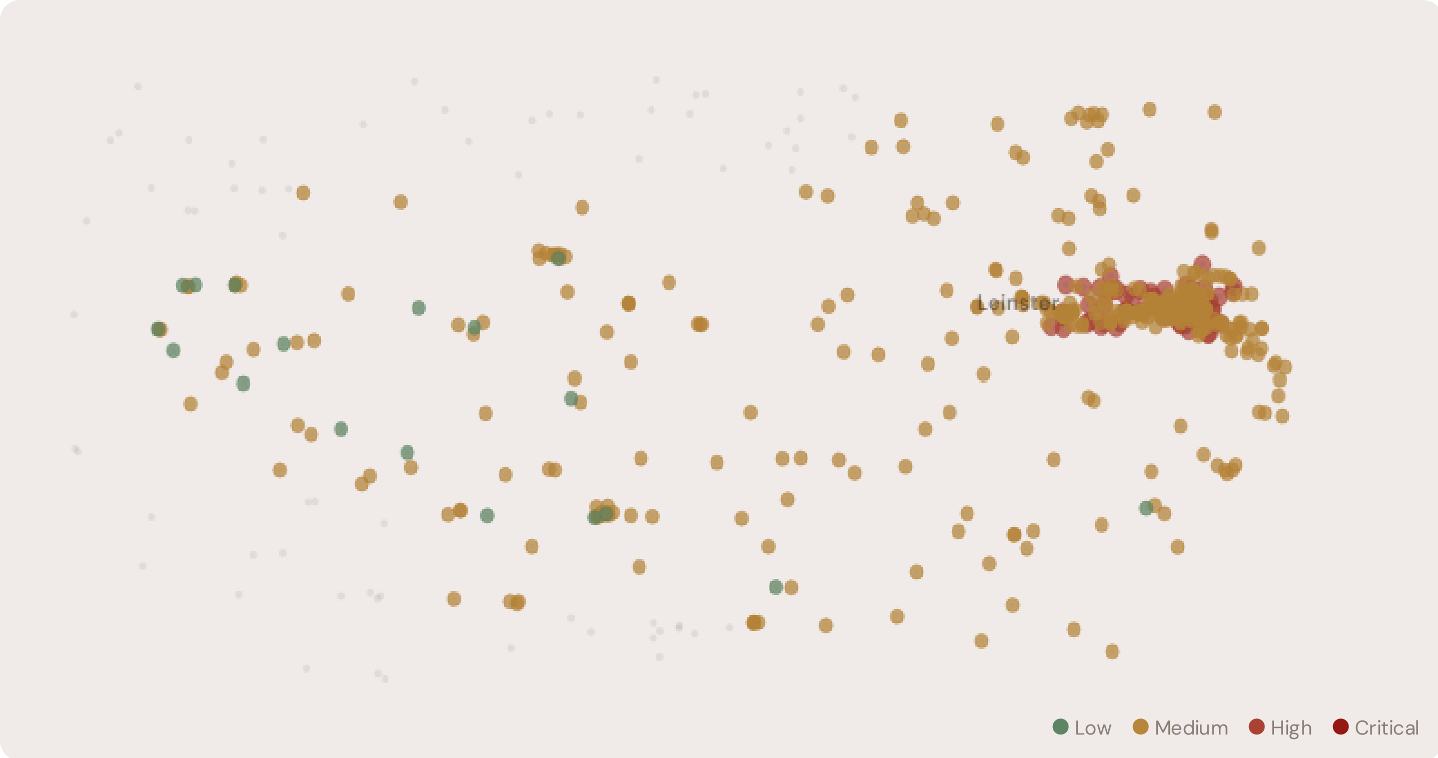
WORST COUNTY

Leinster

Avg R: 0.403 · 549 substations

The Dublin–Cork corridor hosts **549 substations** across six counties, making it one of the most energy-critical corridors in Europe. However, its risk profile is disproportionately concentrated in the upper bands: **87 substations (15.8%)** are classified High, with only **18** in the Low band. 66% of Dublin–Cork corridor substations score above the national fleet median of 0.360. The corridor median R of 0.405 reflects the tension between nuclear baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Climate Action Plan 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
	Leinster	0.642	High	0.700	0.254	0.543	0.423	0.100	0.388	C
Taney 110 kV Substation	Leinster	0.633	High	0.617	0.486	0.491	0.365	0.100	0.469	
Saggart 38kV Substation	Leinster	0.619	High	0.729	0.337	0.490	0.430	0.100	0.477	C
Cheeverstown Luas Substation	Leinster	0.618	High	0.674	0.561	0.431	0.456	0.100	0.454	C
Bedford Row 38kV Substation	Leinster	0.612	High	0.714	0.329	0.412	0.293	0.100	0.380	C
Grange Castle West 110kV Cable Compound	Leinster	0.608	High	0.689	0.293	0.404	0.424	0.100	0.482	C
Irishtown 220-kV Substation	Leinster	0.606	High	0.605	0.499	0.512	0.414	0.100	0.306	
Batter Lane 110 kV Substation	Leinster	0.598	High	0.634	0.583	0.370	0.402	0.100	0.355	
Poppintree 110kV Substation	Leinster	0.592	High	0.725	0.506	0.439	0.315	0.100	0.429	C
Old Bawn 38kV Substation	Leinster	0.585	High	0.622	0.545	0.431	0.379	0.100	0.517	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT

… 539 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — DUBLIN–CORK CORRIDOR VS FLEET

C — Continuity 0.496 vs 0.461 (+8%)

I — Infrastructure 0.376 vs 0.344 (+9%)

S — Saturation 0.100 vs 0.100 (0%)

V — Voltage 0.379 vs 0.349 (+8%)

E — Economic 0.348 vs 0.338 (+3%)

T — Transition 0.382 vs 0.359 (+6%)

MODIFIER AVERAGES — DUBLIN–CORK CORRIDOR VS FLEET

R3 Consequence	1.122	fleet 1.095	↑
R4 Graph Crit.	0.979	fleet 0.978	↓
R6a Restoration	1.019	fleet 1.021	↑
R6b Seismic	1.030	fleet 1.031	↑
R7 Digital Read.	0.979	fleet 0.980	↓

The dominant risk driver across the Dublin–Cork corridor is **T (Transition)**, averaging 0.382 against a fleet average of 0.359 — occupying 764% of its weight ceiling (w = 0.05). The second contributor is **V (Voltage)** at 0.379 vs fleet 0.349 (379% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.122 (fleet: 1.095), reflecting the socio-economic fragility of The Dublin–Cork Corridor's service-oriented economy with significant wind energy integration and growing data centre demand. The R6b seismic modifier averages 1.030, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 County Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Leinster 549 subs · avg R = 0.403 · H:87 M:444

The province-level breakdown reveals significant intra-regional variation. **Leinster** leads with a mean R of 0.403 across 549 substations, while **Leinster** scores 0.403 — a spread of 0.000 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Dublin–Cork corridor is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

0 Dublin–Cork corridor substations score R ≥ 0.650, placing them in the upper quartile of national risk. For Climate Action Plan investment prioritisation, this analysis identifies the Leinster corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

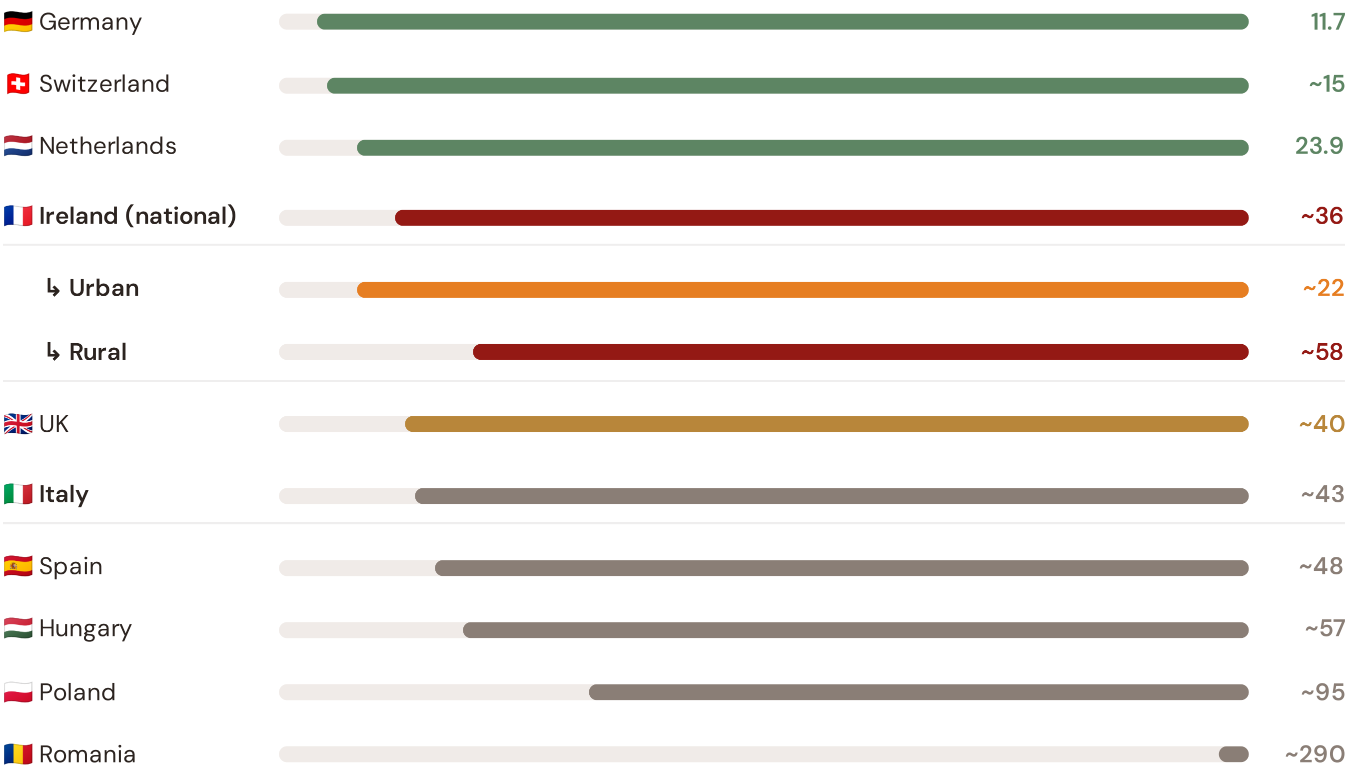
European Context Card

> **How this works:** Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Dublin–Cork corridor analysis hinges on Ireland's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); CRU CRU-Quality (2023); Bundesnetzagentur (2024). Ireland sub-categories from CRU territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Dublin–Cork corridor deep-dive shows **549 substations** (of 549) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Ireland's rural territory classification. The Dublin–Cork Corridor's average C of 0.496 is **1.1× the fleet average** of 0.461. In European terms, these substations individually perform closer to Hungarian or Spanish SAIDI levels (~48–57 min/customer/year) than to Ireland's own national average (~36 min). The SSI reveals what aggregate CEER statistics conceal: that Ireland's mid-tier European position is not a national condition but a statistical average of Dutch-quality urban performance and Eastern-European-level rural performance — and the Dublin–Cork corridor concentrates the worst of the latter.

Regional Risk Ranking — All 26 Irish Counties

Where does the Dublin–Cork corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

#23 Cavan 30 subs · avg R = 0.257 · H:0

#24 Roscommon 25 subs · avg R = 0.255 · H:0

#25 Monaghan 7 subs · avg R = 0.241 · H:0

#26 Leitrim 6 subs · avg R = 0.234 · H:0

The Dublin–Cork corridor ranks **#1 of 26 counties** by mean R_median, placing it among the upper tier of national risk. The three most stressed regions are Dublin, Cork, Limerick, while the three most resilient are Leitrim, Monaghan, Roscommon. 2 of 26 counties score above the fleet average of 0.369. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across Ireland. It is precisely this substation-level granularity that positions the SSI as a tool for Climate Action Plan / NECP investment prioritisation: not treating entire counties as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the ESB Networks/EirGrid and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	ESB Networks/EirGrid + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a €10M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against CRU (Commission for Regulation of Utilities)'s published CRU-Quality data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall visualisation: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The Irish training set comprises 994 substations with complete 95-feature vectors. Average CI_width of 0.055 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 67 Low, 839 Medium, 88 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Celtic Sea Offshore Wind Nexus

Deep-dive into Munster's grid risk profile — substation map, component analysis, province breakdown. European Context Card: Grid readiness for Celtic Sea offshore wind farms and southern interconnection.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

SEAI, Enterprise Ireland are due for refresh in Q3. Impact: ESBN-Data (DER installation register) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CRU (Commission for Regulation of Utilities)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of June 2026. Deep-dive region: **Munster**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Irish utility, CRU, EirGrid, ESB Networks, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 994 substations · 2,306 power lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5
